X1-L1 lab: Capstone 1: refactor the NAT pipeline on your laptop

Platform: Laptop

You need Java 17 or 21 and the `spark` dependency group (see practice/README.md). The acceptance tests run on a local SparkSession with Delta Lake.

1. Read practice/capstone/before_pipeline.py and list every defect you see. Compare your list with the concept section.
2. Read practice/tests/test_x1_pipeline.py. Write down, in one line each, what the 11 tests require.
3. Run `pytest practice/tests/test_x1_pipeline.py`. Every test fails with NotImplementedError. That is your starting point.
4. Implement load_config and PipelineConfig.**post_init**. Run the two config tests until they pass.
5. Implement ingest_bronze, then transform_silver and upsert_silver, then compute_gold. Run the tests after each function.
6. Implement run_checks and enforce, then build_run_record, log_run and get_unprocessed_waves.
7. Run `ruff check practice` and `mypy`. Fix every finding.
8. When all 11 tests pass, compare your module with practice/solutions/x1_pipeline.py. Note two things the reference does differently and why.

Notes